A small bootstrap below the 7/8 quasi-Riemann bound

Research note prepared with OpenAI Codex

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Abstract

We extend the permitted parameter range in the fourth-moment lemma of the OpenAI cubic-theta argument. The printed proof permits $13/18 \le \kappa \le 1$, rather than only $3/4 \le \kappa \le 1$. Combining this observation with a rational reparametrization of the original proof, followed by a second finite bootstrap, yields the zero-free boundary $43747851/50000000 = 0.87495702 < 7/8$. All new exponent inequalities are checked with exact rational arithmetic. The argument imports the analytic lemmas of the original paper; this note is not an independent reproof of those lemmas. The extended energy induction is verified in Lean; the complete improved zero-free theorem is not. Its small improvement should be distinguished from the substantial, still unproved mean-value improvements required for a boundary such as $6/7$.

1 Statement and dependencies

Let $F = \mathbb{Q}(\sqrt{-3})$, and let β_* be the supremum of $1/2$ and the real parts of the nontrivial zeros of all primitive finite-order Hecke L-functions over F, with arbitrary conductor and height. Principal poles are excluded. We use the arithmetic conventions, original zero extensions, fixed excluded prime sets, physical row family, and smooth weights of the OpenAI paper [1]. In particular, no coprimality mask is silently deleted.

Theorem 1. *With the analytic inputs proved in [1],*

$$\beta_* \le B_2 := \frac{43747851}{50000000} = 0.87495702.$$

Consequently every finite-order Hecke L-function over F and every Dirichlet L-function, including the Riemann zeta function, is zero-free in $\Re s > B_2$. The principal pole at $s = 1$ is allowed.

The new analytic point is Lemma 1. The other changes are extensions of parameter-dependent estimates already proved in the source, and exact finite inequalities. We import the following source results, identified by their stable LaTeX labels:

- the 7/8 theorem and the continuation criterion `lem:continuation-criterion`;
- the arithmetic identities, theta reflection, reflected energy `lem:reflected-energy`, and additive Gram bound `prop:probe-gram`;
- the zero detector `prop:detector-witness`, marked inverse moment `lem:marked`, and sixth-power amplification `lem:inverse-amplification`;
- the proof of `lem:plain`, with the range extension established below, and logarithmic control `lem:logarithmic-control`;
- the exact high representation, dynamic local errors `prop:probe-errors`, shared contour lemmas, and late height closure;

- the transfer `prop:hecke-dirichlet-transfer`.

The local source inspected for this note is `formalize_11_12/papers/paper78.tex`. All references to source line numbers below refer to that file. Ordinary theorem dependencies on these inputs are retained. No assertion is made that checking the finite certificates independently verifies the entire original paper.

2 Extending the fourth-moment parameter

Recall the source's fourth-moment estimate. Write M for the effective row width, n_1, n_2 for the two plain-polynomial lengths, z for the total selected prime length, and $A = n_1 + n_2 + z$. The source proves

$$\sum_{k \in \mathcal{R}_z} |S_{\psi_k}(n_1; W_1) S_{\psi_k}(n_2; W_2) Q_{\psi_k}|^2 \ll Z^{M+\epsilon}$$

when $z = 0$, without a plain-length restriction, or when

$$z > 0, \qquad A + (6\kappa - 1)z \le M. \tag{1}$$

For $\kappa < 1$ its hypothesis is $\beta_* \le (1+\kappa)/2$. The prime supports are disjoint, their coefficients belong to the fixed finite ray-character group, and all the original row, mask, smoothness, and height hypotheses of `lem:plain` apply. Its slot mesh is chosen before the slot count. We do not change those hypotheses.

Lemma 1. *The entire fourth-moment statement, including its mesh and uniformity conclusions, remains valid for* $13/18 \le \kappa \le 1$.

Proof. Audit the proof at source lines 12531–14986. Its prime estimate uses logarithmic control on $\Re s = (1+\kappa)/2 + e$ and gives $|Q|^2 \ll Z^{\kappa z + \epsilon}$ with the stated seminorm and height factors. Logarithmic control is uniform for its real parameter in $[1/2, 1]$; here $(1+\kappa)/2 \ge 31/36$. The zero-free hypothesis is unchanged. Thus this step does not require $\kappa \ge 3/4$.

At the terminal width $M \le \rho$, equation (1) gives

$$z \le \frac{M}{6\kappa} \le \frac{3M}{13}, \qquad \kappa z \le M/6.$$

This replaces the printed intermediate bound $z \le 2M/9$ at lines 12932–12933. Only the unchanged bound $\kappa z \le M/6$ is used in the terminal exponent, so its extra loss remains at most $\rho/6$.

For $A > 5M/6$, the centered comparison at lines 12947–12979 uses

$$(6\kappa - 1)z \le M - A < M/6.$$

Since $6\kappa - 1 \ge 10/3$, it follows directly that $z < M/20$. This replaces the printed chain $z < M/21 < M/20$. Put $L = M/4$ as in the source. Then $A - z > 2L$, and its reflected comparison has

$$A_{\mathrm{comp}} \le 3M/2 - A + 2z + \xi \le 23M/30 + \xi,$$
$$A_{\mathrm{comp}} + (6\kappa - 1)z \le 5M/2 - 2A + 2z + \xi \le 14M/15 + \xi.$$

Both required comparison margins remain at least $M/15$, so the original requirement $\xi \le \rho/30$ still suffices.

All remaining parameter uses are unchanged. In the child affine defect, $6\kappa - 1 \ge 0$ permits replacing a surviving length by its parent length. Removing a greedy prefix of whole slots costs

$$\kappa d_z \le F_{\mathrm{act}}/6 + \kappa\eta \le F_{\mathrm{act}}/6 + \eta.$$

The denominator 6κ is bounded away from zero. The aggregate error bound uses only $0 \le 6\kappa - 1 \le 5$. Thus the fixed induction depth, strict width drop, finite mask treatment, exceptional-row estimates, and smooth-seminorm propagation keep the same loss allocations. These are every occurrence of κ in the proof. The uniform assertions at lines 12898 and 14889 consequently hold on the enlarged compact interval as well. □

3 The improved row-count formula

Fix $13/18 \le \kappa \le 3/4$ with $\beta_* \le (1+\kappa)/2$, and put

$$c = \frac{1}{6\kappa}, \qquad \alpha = \frac{5}{6}, \qquad \frac{2}{9} \le c \le \frac{3}{13}.$$

For a zero class write $\delta = 2a - 1$ and let $x = q/\delta \in [0, 1/2]$ be the normalized mean prime amplitude. The floor class $\delta_0 = 1/50$ is handled separately. The source's detector has inverse and plain lengths r, m with $r + m \ge t$, $1 \le t \le 3/2$, up to arbitrarily small requested losses. At row width one the exact plain capacity is now $z_P(m) = c(1 - 2m)$; the inverse capacity stays $z_M(r) = (1 - r)/2$.

Lemma 2. *Suppose the physical available prime length exceeds $7/37$ by a fixed positive amount. Define*

$$D_x = 3 - (1 + 4c)x, \qquad P_x = (2 - 4cx)(1 - x),$$
$$J = (\alpha - \delta)D_x + \delta P_x,$$
$$R_*(\delta, x) = 1 - \delta + \frac{(\alpha - \delta)\delta P_x}{2J}.$$

For $1/50 < \delta \le \alpha$, the source's selected-row argument gives $\#\mathcal{B} \ll U^{R_*+\epsilon}(1 + T_1)^{A_\eta}$. All capacity, binning, and witness losses may be made arbitrarily small before the target is fixed.

Proof. For short witnesses the two ideal exponents are

$$A_I(r) = 1 - \delta[x + (1 - x)r], \qquad S_t(r) = 1 - \delta[2cx + (2 - 4cx)(t - r)].$$

They cross at

$$r_*(t) = \frac{(2 - 4cx)t + (2c - 1)x}{D_x}.$$

Here $D_x \ge 53/26$, $r_*(3/2) = 1$, and

$$r_*(1) \ge \frac{3 - 2c}{5 - 4c} \ge \frac{23}{37}, \qquad \frac{1}{3} \le t - r_*(t) \le \frac{1}{2}.$$

Indeed $r_*(1)$ decreases with x, and its value at $x = 1/2$ increases with c; the complementary length increases with t, and at $t = 1$ its value $(1 - 2cx)/D_x$ increases with x.

Thus inverse capacity is at most $7/37$, while the second marked-width margin is at least $2r - 1 \ge 9/37$. Plain capacity is at most $c/3 \le 1/13 < 7/37$. The original selection, whole-slot rounding, zero-capacity treatment, and height estimates therefore apply. The short crossing value is

$$R_{\mathrm{short}}(t) = 1 - \delta + \frac{\delta P_x}{D_x}(3/2 - t).$$

Long inverse witnesses retain the exponent $L(t) = 1 - \delta + (\alpha - \delta)(t - 1)$. Choose $t = 1 + \delta P_x/(2J)$; it lies in $[1, 3/2]$ and balances the two values, giving R_*. Positivity of J follows from $P_x \ge 10/13$, $D_x > P_x$, and $0 \le \delta \le \alpha$. The boundary $\delta = \alpha$ uses the same no-slot amplified estimate as in the source. Since κ is fixed from an already established bound, no $\Delta/4$ capacity replacement penalty is present. □

4 Changing the probe geometry

Set

$$M = 1 - \ell, \qquad l_x = (M - b)/2, \quad l_y = (M + b)/2, \qquad h = 1 - l_x + \ell. \tag{2}$$

Use exactly the source's finite compensated probe, with total slot length ℓ, and with $X = Z^{l_x}$, $Y = Z^{l_y}$. The slots retain their original physical supports in every rescaled summand. Its signal exponent is

$$C(s) = s + l_x/2 - 1 + h/6 = s - (4 + b)/6.$$

Lemma 3. *If $0 < \ell \le 1/5$, $b > 0$, $l_x > \ell$, $l_y > \ell$, $M > 2\ell$, and $l_y - \ell - 11b/6 > 0$, the normalized probe satisfies*

$$|J_\eta(Z)| \ll_{\eta,\epsilon} Z^{C(B)+\epsilon}, \qquad B = 11/12 - \ell/4.$$

Proof. For a rescaled subset of total length $d \in [0, \ell]$, put $M' = M - 2d$ and $\ell' = \ell - d$. The reflected-energy calculation in the proof of `lem:probe-row-norm` gives

$$E_{\mathrm{ref}} \le M' + ((1 + 3\ell' - 2M')/4)_+ + o(1) = M' + ((5\ell - 1 + d)/4)_+ + o(1).$$

Specifically, the source estimate $T_d \le H - 3d + o(1)$ uses only $M + \ell = 1$; it therefore still makes the maximum in the reflected energy equal to $H + o(1)$. Its two remaining branches give the displayed bound. One must retain this extra energy term: it need not vanish for every rescaled subset.

The Gram hypothesis makes $P_a^2/Y' \ll P_a^{1/6}$ and hence supplies the same additive norm as in the source. Cauchy–Schwarz, tuple counting, and the rescaling coefficient then give an extra exponent

$$f(d) = -d + (5\ell - 1 + d)_+/8.$$

Since $\ell \le 1/5$, $f(0) = 0$, and both of its linear slopes, -1 and $-7/8$, are negative. Thus $f(d) \le 0$ everywhere. The modified probe is bounded by $Z^{l_x/2 + b/12 + \epsilon}$. Its principal normalizer has only a subpower reciprocal, as in the source, and $l_x/2 + b/12 = C(11/12 - \ell/4)$. □

For a target $B \in [437/500, 7/8]$, the source's second Euler region may be extended to $\Re s \ge B$, retaining $\Re w \ge 19/20$ and $\Re z \ge 33/200$. Its good-prime defect exponents $4 - 6\Re s - 6\Re z$ and $1 - \Re s - \Re w - 6\Re z$ remain strictly below -1. All local denominators remain uniformly nonzero. The four principal error exponents

$$-\Re s, \quad -6\Re z, \quad 4 - 5\Re s - 6\Re z, \quad 1 - \Re w - 6\Re z$$

are at most $-B$. Thus the principal slot multiplier is $-1 + O(q_p^{-B})$, and its approximation margin may be any $0 < \mu < B \min_i \ell_i$. The unselected principal Euler correction is holomorphic and can be made within $1/2$ of one by enlarging the fixed prime cutoff. The same principal normalization and residues apply.

The shared contour statements formally use $\sigma_0 \ge 7/8$. Their proofs apply at $\sigma_0 = B$ after this Euler extension. On the small-row line replace their $\mathcal{D}_1(3/8)$ by $\mathcal{D}_1(\epsilon_0)$, where $0 < \epsilon_0 \le B - 1/2$ is fixed. No contour crosses a new pole. The absolute local estimates remain $G_p = O(1)$ off the row and $G_p \ll q_p^{1/2}$ on its support. The principal residue margins remain $l_y/20$ and $h/600$.

5 Exact endpoint and remaining ranges

The high exponent relative to $C(B)$ is

$$E_B(d; \delta, q, R) = a - B + h(17/50 - 1/6) - a l_y - \ell/2 + q\ell + d(R + \delta/2 - 17/50), \tag{3}$$
$$a = (1 + \delta)/2.$$

At $d = h$, $q = x\delta$, $R = R_*$, it becomes

$$E_B(h) = \frac{(1+\delta)(1-l_y)}{2} - \frac{\ell}{2} + x\delta\ell + h(R_* + \delta/2 - 1/6) - B. \tag{4}$$

Start from the source theorem $\beta_* \le 7/8$. The rational geometry

$$B_0 = \frac{34999}{40000}, \qquad b_0 = \frac{1}{8}, \qquad \ell_0 = \frac{5003}{30000}, \qquad \kappa_0 = \frac{3}{4}$$

gives $\beta_* \le B_0$ by the preceding estimates and the same final continuation argument. At this geometry the exact certificate `certificate.py` proves $E_{B_0}(h) \le -9/100000$ on the full rectangle $[0, 5/6] \times [0, 1/2]$, and verifies all remaining strict margins. This is also the rational extension discussed in [2]. The use of $\kappa_0 = 3/4$ is justified by the already proved $7/8$ bound, so this first step has no circular hypothesis.

Now use $\beta_* \le B_0$ and set

$$\begin{gathered} \kappa_1 = 2B_0 - 1 = \frac{14999}{20000}, \qquad c_1 = \frac{10000}{44997}, \qquad B_1 = \frac{17499141}{20000000}, \\ b_1 = \frac{6170279}{50000000}, \qquad \ell_1 = \frac{2502577}{15000000}, \\ l_{x,1} = \frac{106463393}{300000000}, \quad l_{y,1} = \frac{143485067}{300000000}, \quad h_1 = \frac{81196049}{100000000}. \end{gathered} \tag{5}$$

Here $13/18 < \kappa_1 < 3/4$, so Lemmas 1 and 2 apply. The formula $B_1 = 11/12 - \ell_1/4$ is exact.

Lemma 4. *At (5), on $0 \le \delta \le 5/6$, $0 \le x \le 1/2$,*

$$E_{B_1}(h_1) \le -10^{-8}.$$

All hypotheses of Lemma 3, the strict supply conditions $\ell_1/h_1 > 7/37$ and $5\ell_1 - h_1 > 0$, and the Euler conditions above hold. The floor and intermediate row ranges have a strict saving relative to $C(B_1)$. The source's absolute small-row estimate has its own strict saving relative to $C(\beta_*)$.

Proof. This is a finite rational calculation, implemented in `research_audit/bootstrap_certificate.py`. To specify the nonlinear certificate completely, form the bivariate polynomial

$$\begin{aligned} N(\delta, x) = J\big[B_1 - (1+\delta)(1-l_{y,1})/2 + \ell_1/2 - x\delta\ell_1 - h_1(5/6 - \delta/2) - 10^{-8}\big] \\ - \frac{h_1}{2}(\alpha - \delta)\delta P_x. \end{aligned}$$

Thus $N = J(-E_{B_1}(h_1) - 10^{-8})$ exactly. The checker proves $J \ge 174985/269982 > 0$. It partitions the rectangle into 24 rational subrectangles and verifies that every Bernstein coefficient of N on each is nonnegative. The Bernstein basis is nonnegative and sums to one, so these checks prove $N \ge 0$ everywhere. All subdivision endpoints, coefficients, and sign decisions use rational arithmetic; no sampled positivity or floating-point optimization is used for this inference. The source code can emit the entire list of rectangles and exact lower coefficients as a separate certificate.

For completeness, the auxiliary checks include

$$\frac{\ell_1}{h_1} - \frac{7}{37} = \frac{13344541}{819341949} > 0, \qquad 5\ell_1 - h_1 = \frac{6669553}{300000000} > 0.$$

The floor uses $\delta_0 = 1/50$, $q = \delta_0/2$, $R = 1$ and has

$$-E_{B_1}(h_1) = \frac{2939583}{500000000} > 0.$$

For $d \in [1/100, 1/2]$, the source's no-slot choice $t = 1$ gives the common bound $R = 76/75 - 2\delta/3$ including the floor. Its slope in d is positive. The two endpoint values in δ at $d = 1/2$ satisfy

$$-E_{B_1}(1/2; 1/50, 1/100, R) = \frac{2148926943}{10000000000}, \qquad -E_{B_1}(1/2; 5/6, 5/12, R) = \frac{239232781}{90000000000}.$$

Both are positive, and this exponent is affine in δ. For $d \le 1/100$, the absolute outer-row lemma gives, directly relative to $C(\beta_*)$, the saving

$$-\big[h_1(17/50 - 1/6) - l_{y,1}/2 + 63/5000\big] = \frac{1287029401}{15000000000} > 0.$$

The script verifies all other listed length, Gram, residue, and Euler inequalities by direct rational subtraction. □

After proving $\beta_* \le B_1$, perform a second finite bootstrap with

$$\kappa_2 = 2B_1 - 1 = \frac{7499141}{10000000}, \qquad c_2 = \frac{5000000}{22497423}, \qquad B_2 = \frac{43747851}{50000000}, \tag{6}$$
$$b_2 = \frac{24681137}{200000000}, \qquad \ell_2 = \frac{6256447}{37500000},$$
$$l_{x,2} = \frac{425853437}{1200000000}, \quad l_{y,2} = \frac{573940259}{1200000000}, \quad h_2 = \frac{324784289}{400000000}.$$

The same exact checker, with these arguments, gives

$$E_{B_2}(h_2) \le -10^{-9} \quad (0 \le \delta \le 5/6,\ 0 \le x \le 1/2). \tag{7}$$

The polynomial is the one in the proof of Lemma 4, with subscript 1 replaced by 2 and margin 10^{-9}. Its certificate has 26 rational subrectangles, and $J \ge 87487115/134984538 > 0$. The auxiliary inequalities all pass again. For explicit representative margins,

$$\frac{\ell_2}{h_2} - \frac{7}{37} = \frac{587163179}{36051056079} > 0, \qquad 5\ell_2 - h_2 = \frac{26678653}{1200000000} > 0.$$

The floor, smaller intermediate, and small-row savings respectively are

$$\frac{58789949}{10000000000}, \qquad \frac{956906941}{360000000000}, \qquad \frac{5148114961}{60000000000}.$$

Thus the entire conclusion of Lemma 4 holds for stage two as well.

6 Continuation and conclusion

Proof of Theorem 1. First obtain the rational preliminary bound $\beta_* \le B_0$ as described above. We explain the continuation step at either $j = 1$ or $j = 2$, assuming the already proved $\beta_* \le B_{j-1}$. Suppose for contradiction that $\Delta = \beta_* - B_j > 0$. Use the corresponding geometry (5) or (6) and its one fixed value $\kappa_j = 2B_{j-1} - 1$. The zero-free hypothesis of the extended plain moment is justified by $\beta_* \le B_{j-1} = (1 + \kappa_j)/2$, independently of the contradiction. The detector's actual classes satisfy $\delta \le 2\beta_* - 1 \le \kappa_j$; certifying the larger rectangle up to 5/6 is therefore sufficient.

Lemma 3 bounds the normalized physical probe by $Z^{C(B_j)+\epsilon}$. Lemma 4 and its stage-two version control the central and smaller nonprincipal rows. For $1/2 \le d \le h_j$, the frequency slope is positive because

$$R_* + \delta/2 - 17/50 \ge 33/50 - \delta/2 > 0.$$

Thus the bound at h_j bounds the entire interval. Write $m_1 = 10^{-8}$, $m_2 = 10^{-9}$ and choose $0 < \zeta < \min\{5\ell_j - h_j, m_j/8\}$, decreasing it further if needed. Extend to $h_j + \zeta$: slopes are

bounded by two, so its cost is less than $m_j/4$, and $h_j + \zeta < 5\ell_j$ leaves prime supply greater than $1/5 > 7/37$. Rows beyond $Z^{h_j+\zeta}$ are negligible by the source's absolute large-row contour, choosing its fixed z line sufficiently far right after choosing ζ.

Subtract $C(\beta_*)-C(B_j) = \Delta$ from the floor, intermediate, and endpoint exponents computed relative to $C(B_j)$. The absolute small-row lemma already supplies its saving relative to $C(\beta_*)$, so no second subtraction is made there. There is no capacity replacement cost. The order of the remaining choices matters. First choose the moment, capacity, and rounding reserves relative to the fixed high-side margin and Δ, without referring to a principal slot approximation margin. Choose their permitted slot mesh, then an even slot count and disjoint prime windows of total length ℓ_j. Lemma 1 gives uniformity before this slot count is fixed. Now choose $0 < \mu < B_j \min_i \ell_i$; the positive principal contour margins are $l_{y,j}/20$ and $h_j/600$. Only after μ exists choose the remaining amplitude-bin, contour, normalization, and detector losses small relative to it and to the earlier margins and Δ. All these real choices precede the target character. This prevents a circular choice of the mesh through a margin that depends on the later slot count.

For each fixed target choose its allowed arithmetic data and internal seminorm orders. The source's late height closure then chooses the positive height exponent, external tail order, and final threshold in that order. The height cutoff appears only in estimates, not in the physical probe or its principal signal. Consequently, for some $0 < \omega < \Delta$ and $\sigma > 0$ independent of the target,

$$|J_\eta(Z)| \ll_\eta Z^{C(B_j)+\omega}, \qquad |J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-\sigma},$$

where

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s,\eta)}\, ds.$$

The extended Euler analysis makes H_η holomorphic on $\Re s > B_j$ and within $1/2$ of one. These are exactly the source continuation criterion's hypotheses at B_j, contradicting $\beta_* > B_j$. Hence $\beta_* \le B_j$. Apply this finite argument first with $j = 1$ and then with $j = 2$ to obtain $\beta_* \le B_2$.

Restoring finite Euler factors does not introduce zeros in this positive half-plane. For a Dirichlet character χ, the norm-pullback factorization

$$L_F(s, \chi \circ N) = L(s,\chi)L(s,\chi\chi_{-3})$$

then gives the Dirichlet conclusion, with the usual principal poles allowed, exactly as in the source's transfer proposition. □

7 What this result does and does not establish

The gain over $7/8$ is 0.00004298. The bound is also slightly below $(1507 - 2\sqrt{921})/1653 = 0.874957069799\ldots$, the optimum reported for the fixed $\kappa \ge 3/4$ parameter system in [2]. That paper already flags the possibility of a tiny improvement on allowing $\kappa < 3/4$; this note supplies a proof amendment and a conservative exact rational realization of that observation. Prior work by LiXiang [3] also describes this feedback and reports the limiting value $0.874957019420098946\ldots$, slightly better than our conservative rational bound. Its repository explicitly states that the improved Lean conclusion still assumes `ZetaInverse.ArithmeticProbeObligation`; the lower-κ energy theorem is formalized, while the complete physical probe obligations remain open there. We do not import that obligation as a proved theorem: our new analytic proof uses the concrete source reflected-energy, Gram, local-error, and contour lemmas with the amendments given above. No claim of priority is made.

This is an analytic research derivation using the original paper's inputs, with machine-checked exact arithmetic for its new inequalities. A separate Lean overlay verifies the full energy induction with the weaker floor $\kappa \ge 13/18$, retaining the hypothesis $2\beta_* - 1 \le \kappa$. All 63

affected modules compile with `autoImplicit=false`; the exported certified-band and terminal-certificate endpoints depend only on `propext`, `Classical.choice`, and `Quot.sound`. The endpoint is `research_audit/kappa_formal/KappaFormal.lean`; the exact theorem signatures and axiom audit are recorded in `research_audit/kappa_formal/logs/endpoint-axioms.log`. This verifies the energy-induction extension, not the full assembly of the improved moment, physical probe, and zero-free theorem in Lean. Nor does this note independently validate every upstream imported theorem. It proves neither the 6/7 mean-value hypothesis nor GRH. In particular, a lower bound for one exponent model is not an impossibility theorem for all modifications of the cubic-theta method or for future research.

There is nevertheless a useful exact obstruction for the unchanged direct majorant. Write $M = l_x + l_y$, $b = l_y - l_x$, and suppose its Gram and energy exponents satisfy

$$g \geq b/6, \qquad e \geq 0, \quad e \geq (1 + 3\ell - 2M)/4, \quad e \geq M + \ell - 1.$$

The empty-subset low bound then requires $B \geq 5/6 + M/12 - \ell/6 + e/2$. The identity

$$\frac{M}{12} - \frac{\ell}{6} + \frac{e}{2} - \frac{1}{30} = \frac{1}{5}\left[e - \frac{1 + 3\ell - 2M}{4}\right] + \frac{1}{60}\left[e - (M + \ell - 1)\right] + \frac{17}{60}e \geq \qquad 0$$

therefore forces $B \geq 13/15$. This concerns the sufficiency of that particular direct majorant. It already blocks 6/7 and 1/2 within that unchanged majorant, independently of any improvement in the row counts. It does not exclude changing the majorant, retaining cancellations discarded by it, or replacing the physical probe.

The exact barrier including parameter feedback

The preceding 13/15 bound uses only the direct majorant. Retaining also the existing row-count formula gives a stronger, precise limitation. This is a statement about the following sufficient exponent package: retain the direct majorant just displayed, require the high bound (4) for every permitted zero/amplitude class, and use a count exponent $R \geq R_*(\delta, x; c)$ from Lemma 2. Prime-supply restrictions and positive reserves may be retained, since they can only weaken the count. The geometry has $h > 0$, but need not satisfy $M + \ell = 1$ for this calculation. A genuinely stronger count, cancellation between physical rows, or independent exclusion of a critical class is outside this package.

Proposition 1. *Within this exponent package, reparametrization and repeated feedback of proved zero-free bounds into the same fourth moment cannot certify $\theta < \theta_*$, where*

$$\theta_* = \frac{11}{12} - \frac{\lambda_*}{4}, \qquad 657\lambda_*^3 - 954\lambda_*^2 + 21\lambda_* + 20 = 0, \quad \frac{1}{6} < \lambda_* < \frac{1}{5}.$$

The specified root is unique, and exact rational sign evaluations give

$$0.87495701942009894 < \theta_* < 0.87495701942009895.$$

The lower bound is sharp for the three-constraint relaxation in the proof. This proposition does not assert sufficiency at the exact endpoint for every estimate of the full analytic argument.

Proof. The direct majorant already forces $\theta \geq 13/15$, so consider only $13/15 \leq \theta \leq 7/8$. Under a contradiction $\beta_* > \theta$, an available fourth moment with $\beta_* \leq (1 + \kappa)/2$ requires $\kappa > 2\theta - 1$. Consequently its capacity is at most the optimistic limiting value

$$c \leq C(\theta) := \frac{1}{6(2\theta - 1)}, \qquad \frac{2}{9} \leq C(\theta) \leq \frac{5}{22}.$$

The case $\kappa = 1$ also respects this inequality. It holds at each class even if different valid moment parameters are selected by class.

At $x = 1/2$, put $D = 5/2 - 2c$, $P = 1 - c$, $J = \alpha D - \delta(3/2 - c)$. Direct differentiation gives

$$\frac{\partial R_*}{\partial c} = -\frac{\delta(\alpha - \delta)^2}{[\alpha(5 - 4c) + \delta(-3 + 2c)]^2} \le 0.$$

Furthermore

$$R_*(\delta, 1/2; c) - \frac{2}{3} = \frac{Q_c(\delta)}{36J}, \quad Q_c(\delta) = (36 - 18c)\delta^2 + (57c - 78)\delta + 25 - 20c.$$

For $2/9 \le c \le 5/22$, $Q_c(1/3) = 3(1 - c) > 0$, $Q_c(1/2) = 4c - 5 < 0$, and $Q_c'(\delta) \le 39c - 42 < 0$ on $[1/3, 1/2]$. Let $\delta(c)$ be its unique root in that interval and take $\delta = \delta(C(\theta))$. It is an admissible test class: it exceeds the detector floor and is below every possible κ, since $\kappa \ge 2\theta - 1 \ge 11/15 > 1/2$. The optimistic exponent at this class is $2/3$; monotonicity implies every permitted capacity has $R \ge 2/3$ there.

The two relaxed low constraints are

$$\theta \ge L_0 = \frac{5}{6} + \frac{M}{12} - \frac{\ell}{6}, \qquad \theta \ge L_K = \frac{1}{3} + \frac{7M}{12} + \frac{\ell}{3}.$$

Since $h > 0$, substituting $x = 1/2$, $R = 2/3$ in the high bound gives

$$\theta \ge H_\delta = 1 + \delta - \frac{1 + \delta}{2} M + \delta\ell.$$

Its coefficient of b cancels. The positive weights

$$w_0 = \frac{4 + 18\delta}{9 + 18\delta}, \qquad w_K = \frac{2}{9 + 18\delta}, \qquad w_H = \frac{3}{9 + 18\delta}$$

sum to one and cancel both M and ℓ. Hence necessarily

$$\theta \ge \frac{7 + 18\delta}{9 + 18\delta}, \qquad \delta = \delta(C(\theta)).$$

The right-hand function is strictly increasing in δ. Define

$$d(\theta) = \frac{9\theta - 7}{18(1 - \theta)}, \qquad \lambda = \frac{11}{3} - 4\theta.$$

Here $d(\theta) \in [1/3, 7/18]$ and $\lambda \in [1/6, 1/5]$. The necessary inequality is equivalent to $d(\theta) \ge \delta(C(\theta))$, or $Q_{C(\theta)}(d(\theta)) \le 0$. The exact substitution identity is

$$Q_{C(\theta)}(d(\theta)) = \frac{-3P(\lambda)}{(5 - 3\lambda)(1 + 3\lambda)^2}, \qquad P(\lambda) = 657\lambda^3 - 954\lambda^2 + 21\lambda + 20.$$

Its denominator is positive. Thus $P(\lambda) \ge 0$ is necessary. Now $P(1/6) = 1/24 > 0$, $P(1/5) = -1088/125 < 0$, and throughout the interval

$$P'(\lambda) = 1971\lambda^2 - 1908\lambda + 21 \le 1971/25 - 1908/6 + 21 = -5454/25 < 0.$$

There is one root λ_*, and necessity becomes $\lambda \le \lambda_*$, that is $\theta \ge \theta_*$. The equivalent cubic is $7884\theta^3 - 18819\theta^2 + 14643\theta - 3686 = 0$. The derivative identity, dual cancellations, exact substitution multiplier, and rational isolating signs are independently verified by `research_audit/feedback_barrier_certificate.py`, using only exact polynomial and rational arithmetic.

At the limiting capacity and root, the relaxation has equality at $\ell = \lambda_*$, $M = 1 - \lambda_*$: $L_0 = L_K = H_\delta = \theta_*$. This proves the stated sharpness for the relaxation. The scalar λ used to eliminate the target is not an assumption about the free geometric ℓ before this equality case. □

This barrier concerns the uniform sufficient bounds, not the existence of zeros in the critical class used to test them. It rules out further gains from geometry and this parameter feedback alone. It does not rule out stronger mean values, an altered probe, or treatments preserving cancellations discarded by the current majorants.

The supplied Gram-splitting route

The prior note `gram_split/note/theta_split_note.tex` correctly retracts its stronger unconditional claims. Reflection contributes a row phase of the form

$$\zeta_m = (\text{fixed ray factor})\,\mu(m)\overline{\alpha(m)}\,\overline{\gamma_2(m)} \quad (m \text{ squarefree}).$$

Consequently its relevant correlation is a pairing of $\pi = \rho\zeta$ with the reflected row, rather than a pairing of ρ alone. An estimate for the latter sequence cannot be substituted for an estimate of the former. The required saving for the correct sequence, including every marked block if slots are retained, remains unproved in that route. The new bootstrap above does not use that saving.

The norm information alone cannot repair this step. Abstract vectors $u = Q^{5/12}v$ and $w = Q^{1/2}v$, with $\|v\|_2 = 1$, satisfy $\|u\|_2^2 = Q^{5/6}$, $\|w\|_2^2 = Q$, and attain $|\langle u, w\rangle| = Q^{11/12}$. Thus those two bounds alone imply no power saving. Additional arithmetic cancellation is necessary.

Even after removing all additional nonnegative losses, the residual Cauchy–Schwarz template still requires

$$B \geq \frac{5}{6} + \frac{l_x - \ell}{6} + \frac{g + e}{2} > \frac{5}{6} \quad \text{if } l_x > \ell, \quad g, e \geq 0.$$

This is a second, weaker but more robust, obstruction to $1/2$ within that precisely specified residual estimate. It does not forbid treating the whole residual by a new cancellation argument.

The alternate $11/12$ extraction route

In the companion paper [4], a mean square over rows of length D^κ combined with prime sixth-power extraction gives the comparison exponent

$$B_{\text{prime}}(\kappa) = \max\left\{\frac{1}{2} + \frac{5\kappa}{12},\ 1 - \frac{\kappa}{6}\right\}.$$

Even if the required mean square were available at any $\kappa \geq 0$, this extraction has minimum $6/7$, achieved at $\kappa = 6/7$. Indeed the convex combination with weights $2/7$ and $5/7$ of the two displayed affine functions is identically $6/7$. The original theorem supplies the mean square only beyond the diagonal, $\kappa > 1$, yielding $11/12$ in the limiting exponent. Integer sixth-power inversion can avoid the second term, but then approaching $1/2$ requires proving mean-square bounds for arbitrarily short row families. Neither the cited proof nor the present bootstrap establishes those estimates. This is a missing input, not a proof that such estimates are false or equivalent to GRH with the same uniformity.

Finally, the older analysis describes perfect cubic large-sieve orthogonality as an open conjecture. That description is obsolete: [5] proves non-orthogonality for the cubic and quartic families. Its sextic prediction is a conjecture; it cannot itself be used as a proved sextic obstruction or as a cancellation theorem for the specific coefficient sequence π above.

References

[1] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane $\Re s > 7/8$*, 30 September 2026. Local source inspected: `formalize_11_12/papers/paper78.tex`. Public manuscript. Stable source labels are used in this note to distinguish theorem statements from the extensions of their proofs required here.

[2] Baiying Liu, *Slightly Improved Zero-Free Half-Planes for the Quasi-Riemann Hypothesis*, 8 October 2026. Local text: `six_sevenths/lit/liu1.txt` and `six_sevenths/lit/liu2.txt`. https://arxiv.org/abs/2610.12234.

[3] LiXiang, *RH-Zero-Free-Formalization*, source repository and README, https://github.com/lixiang90/RH-Zero-Free-Formalization. The repository distinguishes the formalized lower-parameter energy theorem from the conditional improved-bound theorem.

[4] OpenAI, *The Quasi-Riemann Hypothesis*, alternate 11/12 proof, 5 October 2026. Public manuscript.

[5] A. de Faveri, A. Dunn, and J. Hoffstein, *Non-orthogonality of the cubic and quartic large sieves via Rankin–Selberg*, 2026. https://arxiv.org/abs/2607.07911.